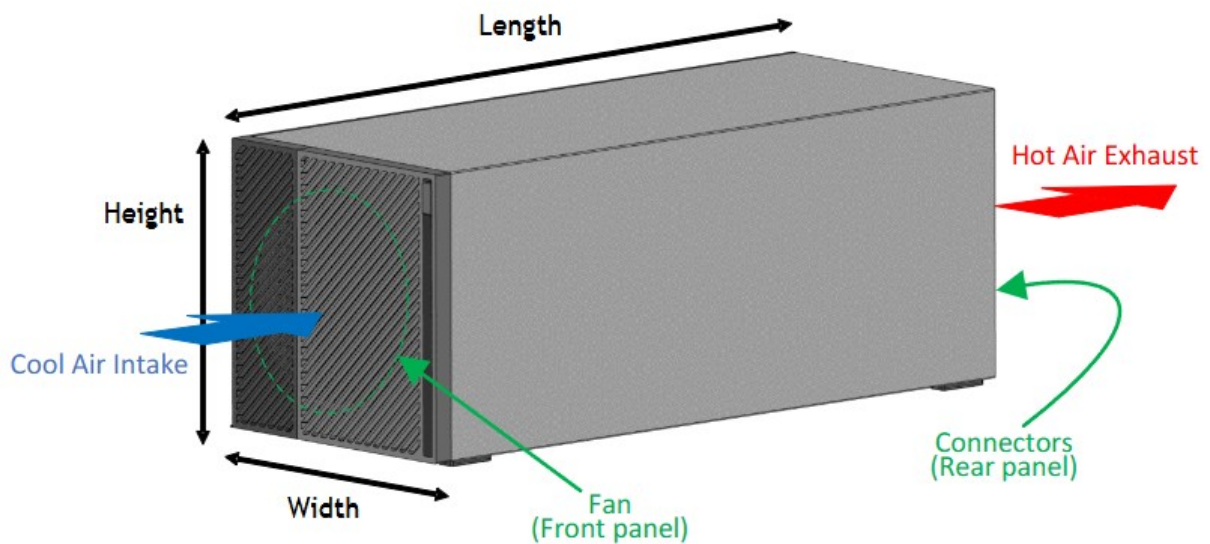


SUMMER RESEARCH PORTFOLIO

Autonomous Driving Data + Embedded Hardware Integration



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Mi3 Laboratory | University of California, Merced | Summer 2026

Projects: doPlan dataset annotation and NVIDIA DRIVE AGX Thor bench integration

Portfolio Overview

A concise record of the research, engineering, and communication work completed during Summer 2026

This portfolio documents two connected contributions completed in Dr. Ross Greer's Mi3 Laboratory: human annotation for the doPlan autonomous-driving dataset and the design, construction, verification, and successful boot of a custom bench power harness for an NVIDIA DRIVE AGX Thor developer system.

From passenger-centered language data to the embedded hardware capable of running advanced vehicle AI workloads.

Major outcomes

- Created passenger-style natural-language annotations for variable-length nuPlan driving clips.
- Applied a four-class referential taxonomy: non-referential, static, dynamic, and combined.
- Learned to distinguish instructions from descriptions and to handle ambiguous long-horizon clips.
- Deciphered NVIDIA mechanical and power documentation for the in-vehicle Thor platform.
- Designed and built a fused 12 V bench power harness with parallel VBAT and ground contacts.
- Verified continuity, isolation, polarity, and output voltage before connecting the developer system.
- Successfully powered and booted the DRIVE AGX Thor to its desktop environment.
- Presented the summer research at the SURI 2026 20th Annual Symposium at UC Merced.

How to use this archive

The ZIP archive is divided into a shareable portfolio and a private source archive. The shareable section is appropriate for internship applications and interviews. The private section contains confidential documentation and collaboration screenshots and should not be redistributed.

Summer at a Glance

Two workstreams, one systems-level story

Workstream	Contribution	Result
doPlan	Watch multi-view driving clips, write passenger instructions, assign referential tags	Contributed labels and practical annotation insights to a long-horizon language-guided driving dataset
DRIVE AGX Thor	Identify power requirements, design and fabricate a bench harness, perform electrical checks	Successfully powered and booted the in-vehicle developer system
Research communication	Develop and deliver an oral presentation with project visuals and a live boot video	Presented at the SURI 2026 20th Annual Symposium; presentation received strong faculty feedback

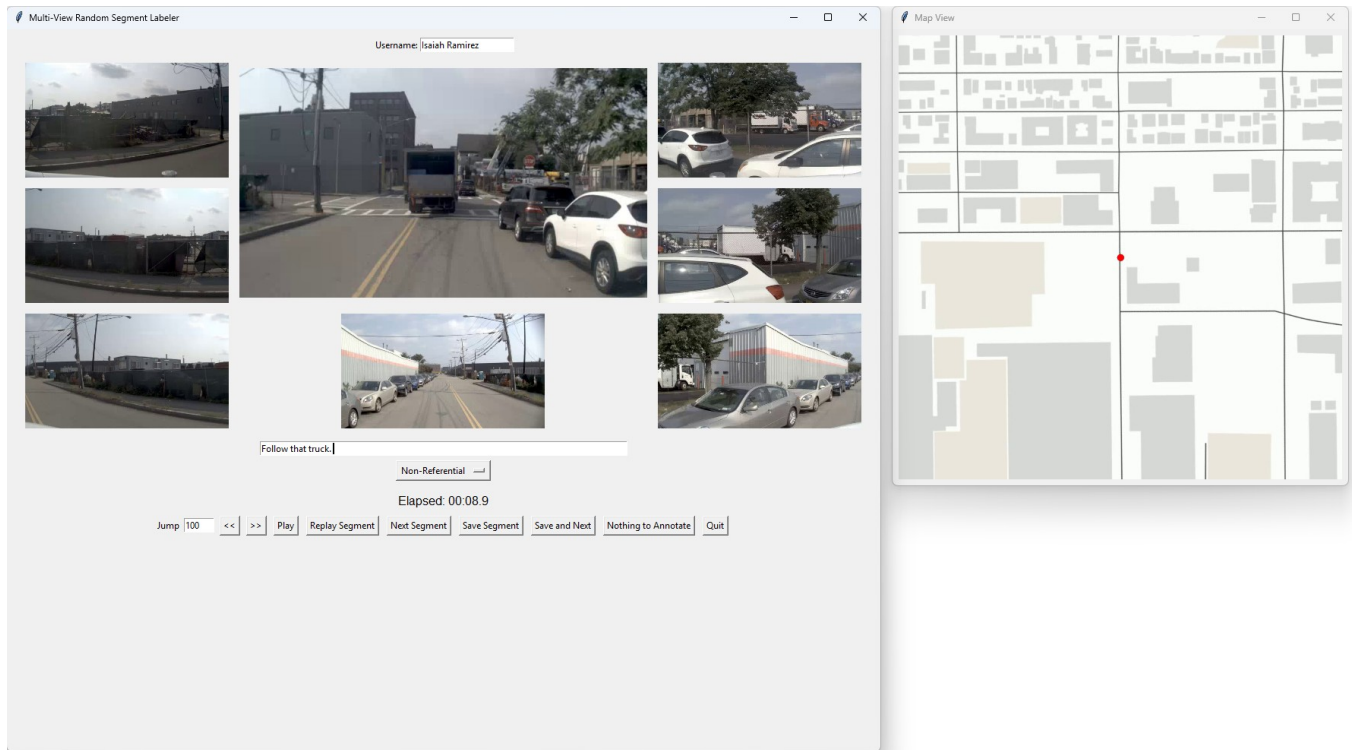
Engineering process used throughout the summer

1. Understand the user or system need
2. Read primary documentation and project guidance
3. Translate requirements into an implementation plan
4. Build or annotate carefully and consistently
5. Verify results with evidence
6. Communicate decisions, limitations, and outcomes

Project 1: doPlan

Annotating long-horizon passenger instructions for autonomous vehicles

doPlan pairs variable-length driving scenes from nuPlan with natural-language instructions written from a passenger perspective. The dataset is intended to support future systems that connect language, perception, and planning over time.



The multi-view annotation interface used during the project. It combines front, side, and rear camera context with a map view and an instruction entry field.

My annotation workflow

- 1. Watch the complete clip** - Observe the vehicle behavior, traffic agents, landmarks, and sequence of actions.
- 2. Think like a passenger** - Ask what a passenger would naturally say to cause the behavior.
- 3. Write an instruction** - Use natural language rather than narrating what the vehicle did.
- 4. Assign a referential class** - Identify whether the instruction depends on no reference, a static feature, a dynamic agent, or both.

Annotation Judgment

Why the work was more than typing labels

Description: "The vehicle turns left." | Instruction: "Turn left."

The central annotation challenge was preserving passenger intent. A description reports observed behavior; an instruction expresses the action a passenger wants the driver to perform. The distinction is subtle, but it determines what a future model learns from the dataset.

Referential classes

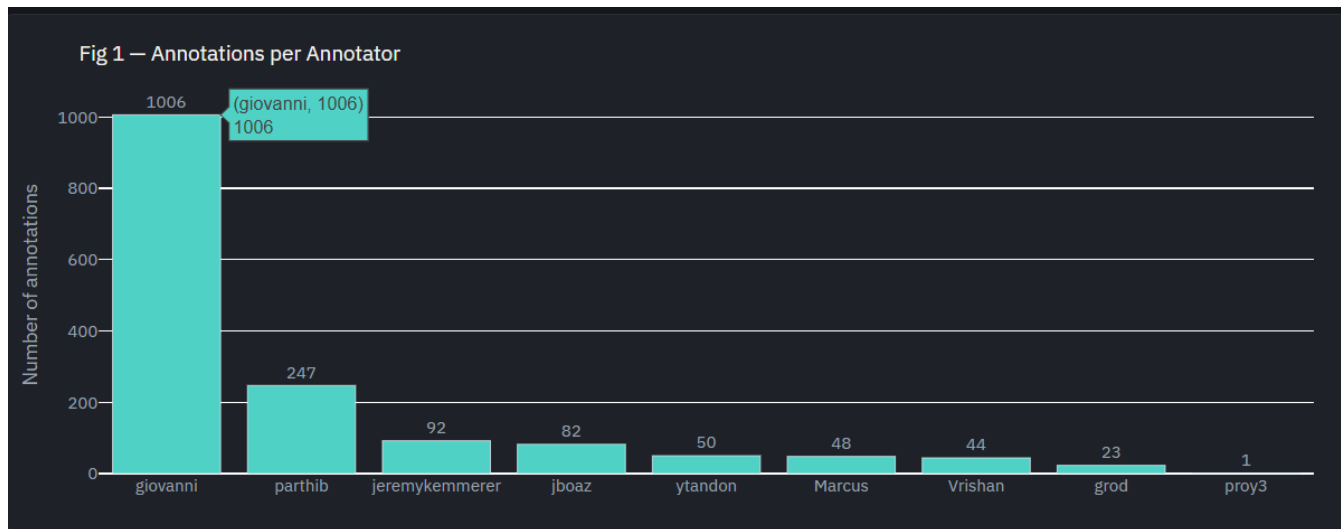
Class	Example	Meaning
Non-referential	Turn left.	No specific object or landmark.
Static	Turn at the stop sign.	Refers to a fixed road feature or landmark.
Dynamic	Go around the truck.	Refers to a vehicle, pedestrian, cyclist, or other moving agent.
Combined	Merge behind the bus near the intersection.	Uses both static and dynamic references.

Difficult cases encountered

- The clip showed several turns, but the final destination was unknown.
- A long clip could be represented as one high-level instruction or multiple turn-by-turn instructions.
- Different annotators could produce different but valid wording for the same behavior.
- Small changes in wording could change the referential category or implied intent.
- When intent was not visible, the annotation needed to remain grounded rather than inventing a destination.

Dataset Progress and Collaboration

A dataset grows through many individual judgment calls



A July 15 project dashboard snapshot showing annotation contributions. This was an intermediate snapshot before all summer researchers submitted their work.

The dashboard shown here recorded approximately 1,600 labels at that point in the summer. It was not the final project total. The project involved approximately 15 annotators overall, and individual annotations were estimated to take roughly two minutes, with difficult scenes taking longer.

Regular lab discussion helped refine the annotation process. Questions included how to save blank segments, whether long clips should use high-level or turn-by-turn instructions, and how interface improvements such as map orientation cues could help annotators.

Contribution evidence

- Used the project annotation interface and submitted labels in the required format.
- Applied lab guidance to ambiguous scenes and adjusted annotation style over time.
- Participated in a broader team effort to expand coverage for downstream research and analysis.
- Explained the annotation methodology, challenges, and significance to a general research audience.

Project 2: NVIDIA DRIVE AGX Thor

Designing a safe bench power solution for an in-vehicle developer system

The laboratory had an in-vehicle NVIDIA DRIVE AGX Thor developer system but did not have the standard bench power adapter. The engineering objective was to understand the power interface, fabricate a custom harness, verify it electrically, and determine whether the system could be powered safely from a laboratory supply.



Rear panel of the DRIVE AGX Thor system.



Mean Well 12 V bench power supply connection used for the completed test.

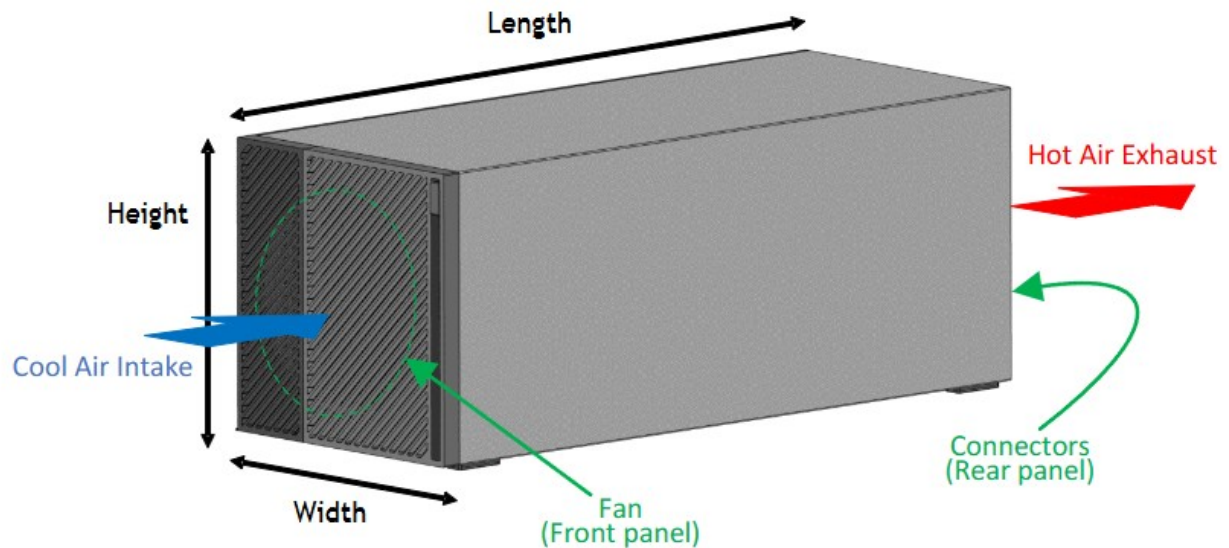
Initial unknowns

- Correct mating connector and crimp contacts
- Power and ground pin assignments
- Required number of parallel conductors
- Appropriate wire gauges and fuse arrangement
- Meaning and necessity of the cable-present signal
- Safe verification procedure before connecting expensive hardware

Documentation and Electrical Architecture

Translating the mechanical guide into a buildable harness

The NVIDIA mechanical documentation identified two DC power headers. For each header, pins 1 through 6 are VBAT supply and pins 7 through 12 are ground. The signal group also includes S2, labeled cable present and active low, with a note that the developer system uses a 4.7 kOhm pull-down. The first power-up was performed without adding the S2 connection, and the system still powered and booted.



Thor mechanical form-factor illustration used in the final presentation.

Final topology: 12 V supply -> inline fuse -> 12 AWG trunk -> six 16 AWG conductors -> Thor power contacts

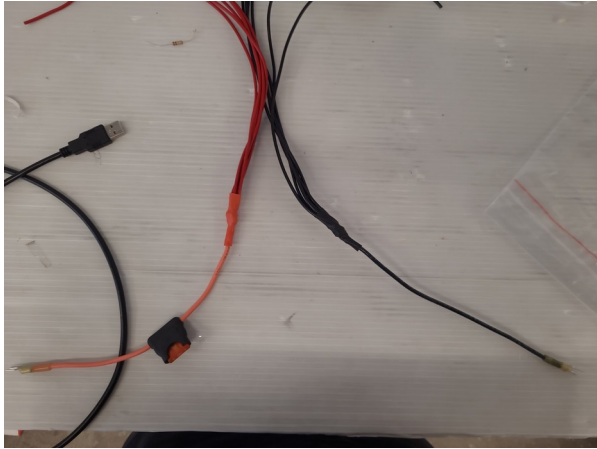
Harness Design Decisions

Electrical capacity, protection, manufacturability, and mechanical reliability

Decision	Reasoning	Implementation
Parallel contacts	Current must be distributed across the connector rather than carried by one contact	Six positive and six ground conductors
12 AWG main leads	Main trunk carries the combined current	Short red and black supply leads
16 AWG branch wires	Fits the selected contacts and provides current margin per branch	Six red and six black branches
Inline fuse	Protects the positive power lead and harness	Heavy-duty 12 AWG fuse holder
Staggered splices	Reduces harness bulk and improves sleeving	Positive and ground transitions offset from each other
Insulation and strain relief	Prevents exposed conductors and supports the splice	Soldered joints and heat shrink

Harness Fabrication

From individual conductors to a finished split-power cable



Positive and ground transitions from one main lead to six branch wires. The two splices were staggered.



The twelve branch wires terminated in the custom Thor connector housing.

Each side of the harness transitions from a single 12 AWG trunk to six 16 AWG wires. The splices were soldered, insulated with heat shrink, and mechanically checked before the contacts were inserted into the connector housing.

Fabrication sequence

- Cut and organize six positive and six ground branch conductors.
- Create and solder the positive and ground fan-out splices.
- Insulate the joints and stagger their positions.
- Crimp the power contacts and perform individual tug tests.
- Insert contacts into the correct connector cavities and verify retention.
- Connect the fused positive trunk and ground trunk to the Mean Well supply.

Verification and First Boot

A staged test process before connecting the Thor

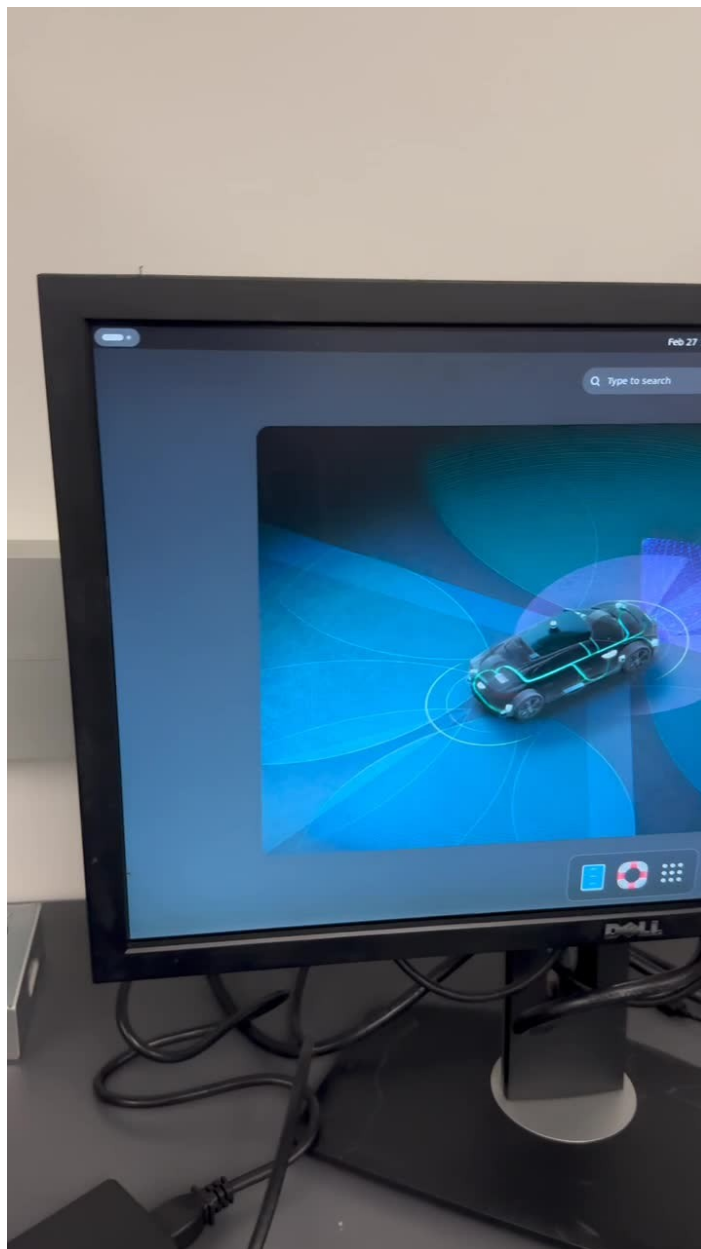
The harness was not connected to the developer system until the electrical checks passed. This reduced the risk of reversed polarity, a short circuit, an open branch, or an incorrect connector assignment.

Pre-connection checks

Test	Acceptance result
Isolation	No continuity between positive and ground bundles.
Positive continuity	Every red connector contact connected to the +V trunk.
Ground continuity	Every black connector contact connected to the -V trunk.
Polarity	Approximately +12 V measured from a ground contact to a positive contact.
Final inspection	Fuse path verified; contacts seated; no exposed strands; splices passed tug testing.

Successful System Boot

Final validation of the custom bench power harness



First-boot video frame showing the Thor desktop.

Result: the custom fused harness successfully powered and booted the NVIDIA DRIVE AGX Thor developer system.

The included six-second video records the powered system at the desktop. This result closed the loop from documentation review and component selection through fabrication, electrical verification, and full system bring-up.

Research Presentation

SURI 2026 20th Annual Symposium - UC Merced - July 31, 2026

The summer concluded with an oral presentation at the Summer Undergraduate Research Institute symposium. The presentation explained the motivation for passenger-centered driving instructions, the doPlan annotation workflow, referential classes, annotation challenges, dataset progress, and the completed Thor harness project.



Symposium identification badge from the SURI 2026 20th Annual Symposium at the University of California, Merced.

Presentation outcomes

- Delivered a roughly ten-minute oral presentation for a multidisciplinary audience.
- Used original screenshots, diagrams, examples, and an autoplay video of the Thor boot.
- Connected the data-annotation work to the embedded computing hardware used in autonomous-driving research.
- Responded to questions about the datasets, annotation ambiguity, and bench integration.
- Received positive feedback from Dr. Ross Greer, who specifically encouraged inclusion of the Thor engineering result.

Photo placeholder: add photographs of Isaiah presenting, the symposium venue, and the research group to the Symposium media folder.

Skills Demonstrated

Evidence that can be discussed in interviews

Skill	Evidence	Interview framing
Research annotation	Created passenger instructions and referential labels	I translated ambiguous driving behavior into structured training data.
Technical documentation	Interpreted NVIDIA mechanical and quick-start guides	I extracted electrical requirements from incomplete and specialized documentation.
Harness design	Selected conductors, contacts, fuse, and topology	I designed the complete power path and distributed current across parallel contacts.
Fabrication	Crimping, soldering, heat-shrink insulation, connector assembly	I converted a design into a mechanically robust physical harness.
Electrical verification	Continuity, isolation, polarity, voltage, and mechanical checks	I used a staged validation plan before energizing expensive equipment.
Embedded-system bring-up	Powered and booted DRIVE AGX Thor	I completed a hardware integration task through successful system boot.
Communication	Symposium presentation and Q&A	I explained technical work clearly to a mixed audience.
Collaboration	Worked within the Mi3 research team	I incorporated evolving guidance and communicated progress and uncertainty.

Resume and Portfolio Language

Concise descriptions for applications

Research experience sentence

Research Apprenticeship with Dr. Ross Greer, Computer Science & Engineering, UC Merced. Project: doPlan - Annotating Long-Horizon Driving Instructions, Summer 2026: created passenger-centered driving annotations, applied referential taxonomies, and engineered a custom NVIDIA DRIVE AGX Thor bench power harness.

Resume bullets

- Annotated variable-length autonomous-driving clips with passenger-centered natural-language instructions and referential classes for the doPlan research dataset.
- Interpreted NVIDIA DRIVE AGX Thor mechanical and installation documentation to identify the in-vehicle DC power interface and bench-integration requirements.
- Designed, fabricated, and verified a fused 12 V power harness using a 12 AWG main trunk and twelve 16 AWG connector branches.
- Performed continuity, isolation, polarity, voltage, and mechanical retention checks before successfully powering and booting the DRIVE AGX Thor system.
- Presented summer research at the SURI 2026 20th Annual Symposium at UC Merced and communicated results to a multidisciplinary audience.

Thirty-second project summary

During Summer 2026, I worked in UC Merced's Mi3 Lab on doPlan, a dataset for teaching autonomous vehicles to understand passenger instructions. I created language annotations and referential labels, then completed a separate hardware-integration task by designing and building a custom fused bench power harness for an in-vehicle NVIDIA DRIVE AGX Thor. After staged electrical verification, the system successfully booted to the desktop. I presented both projects at UC Merced's SURI symposium.

Reflection

What changed during the summer

The doPlan work showed that data collection is an engineering activity: prompts, interfaces, categories, and human judgment shape what a future model can learn. The Thor work showed the other side of the system: advanced AI is only useful when the hardware, power, interfaces, and validation process are correct.

Key lessons

- Ambiguity should be acknowledged and managed rather than hidden.
- Primary documentation is essential, but it may not answer every integration question directly.
- High-current wiring requires attention to both electrical capacity and mechanical reliability.
- Testing should proceed from low-risk checks to full system operation.
- Clear communication can make a technical contribution visible and understandable.
- The intersection of embedded systems, computer architecture, FPGA design, and intelligent vehicles is a strong long-term area of interest.

Future directions

- Continue developing FPGA and hardware/software co-design skills.
- Pursue a senior project involving an FPGA-based graphics accelerator for embedded Linux.
- Seek paid internships involving embedded systems, digital design, hardware integration, or autonomous systems.
- Maintain the Mi3 collaboration when time, location, and funding make participation practical.